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A Comparative Study of Machine Learning Algorithms for Predicting Customer Churn: Analyzing Sequential, Random Forest, and Decision Tree Classifier Models

1st Ratnam Dodda
Department of CSE (AI&ML)
CVR College of Engineering
Hyderabad, India
ratnam.dodda@gmail.com

2nd Dr. C. Raghavendra
Department of CSE (DS)
CVR College of Engineering
Hyderabad, India
drcraghavendra@gmail.com

3rd Marpu Aashritha
Department of Computer Science and Information Technology
CVR College of Engineering
Hyderabad, India
aashrithamarpu17@gmail.com

4th Hima Varshini Macherla
Department of Computer Science and Information Technology
CVR College of Engineering
Hyderabad, India
frooty.macherla7@gmail.com

5th Anusha Reddy Kuntla
Department of Computer Science and Information Technology
CVR College of Engineering
Hyderabad, India
anushakuntla79@gmail.com

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***Abstract*—Predicting customer churn is vital for businesses aiming to retain their customer base and enhance profitability. This study investigates the effectiveness of three prominent machine learning algorithms Sequential, Random Forest, and Decision Tree Classifier in forecasting customer churn. Utilizing a dataset encompassing historical customer information such as demographics, transaction records, and interaction patterns, we preprocess the data by addressing missing values, encoding categorical variables, and scaling numerical features. Subsequently, each model is trained on the processed dataset, and their performance is assessed using accuracy as the evaluation metric. Our findings demonstrate promising predictive capabilities, with the Sequential model achieving an accuracy score of 94%, followed by Random Forest with 92%, and Decision Tree Classifier with 91%. Additionally, we conduct a comparative analysis of these models, focusing on interpretability, computational complexity, and robustness. While the Sequential model excels in predictive accuracy by capturing temporal dependencies in sequential data, Random Forest and Decision Tree Classifiers offer simplicity and interpretability, catering to scenarios where model transparency is paramount.**

***Index Terms*—Customer Churn Prediction, Machine Learning Algorithms, Sequential Model, Random Forest, Decision Tree Classifier, Comparative Analysis**

I. Introduction

Customer churn prediction: Customer churn prediction refers to the task of forecasting which customers are likely to terminate their relationship with a business in the near future. It involves analyzing historical customer data, such as purchase behavior, engagement metrics, and demographic information, to identify patterns and indicators that precede churn events. By leveraging advanced deep learning algorithms, businesses can proactively intervene with targeted retention strategies, thereby reducing churn rates and maximizing customer lifetime value [1].

Application: The application of deep learning in customer churn prediction spans across various industries, including telecommunications, banking, e-commerce, subscription services, and more. For example, telecom companies use churn prediction models to identify customers likely to switch to a competitor, allowing them to offer personalized incentives or service upgrades to retain these customers. Similarly, e-commerce platforms utilize churn prediction to anticipate customer attrition and tailor marketing campaigns to re-engage at-risk customers [2].

Objectives: The objective of customer churn prediction using deep learning is to leverage advanced computational techniques to accurately anticipate when customers are likely to discontinue their relationship with a business or service. By employing deep learning algorithms, which are capable of processing large volumes of complex data, the aim is to identify patterns and trends indicative of potential churn behavior. Through this predictive modeling approach, businesses can proactively intervene and implement targeted retention strategies to mitigate customer attrition, thereby preserving revenue streams and maintaining a loyal customer base [3].

Motivation: The motivation behind this study stems from the increasing recognition of the importance of customer retention in driving sustainable business growth. By proactively identifying and addressing churn risks, companies can not only preserve existing revenue streams but also cultivate stronger customer relationships and enhance brand loyalty. Furthermore, with the proliferation of data analytics and machine learning technologies, businesses have unprecedented opportunities to extract actionable insights from vast amounts of customer data, thereby gaining a competitive edge in the marketplace [4].

Organization of Report: The report on customer churn prediction using deep learning can be organized into several key sections. Firstly, an introduction should provide background information on customer churn, its significance to businesses, and the potential of deep learning in predictive analytics [3]. Following this, the methodology section should detail the deep learning architecture employed, including the selection of neural network models, data preprocessing techniques, and evaluation metrics utilized [5]. The data collection and preprocessing section should outline the sources of data, data cleaning procedures, and feature engineering methods applied to prepare the dataset for model training. Subsequently, the results and analysis section should present the performance metrics of the deep learning models, including accuracy, precision, recall, and F1-score, along with any insights gained from the analysis of model predictions. Furthermore, a discussion section should delve into the implications of the findings, potential limitations of the study, and avenues for future research [6].

A. Literature Review

In their seminal work, Saran Kumar A and Chandrakala D (2016) [7]highlighted the exponential growth in data volume over the past two decades due to technological advancements. This surge in data necessitated the development of various processes and methodologies to effectively handle and extract valuable insights from raw data, a practice commonly referred to as data mining. Across various industries, data mining techniques have proven instrumental in uncovering hidden patterns and valuable information. Of paramount importance to businesses is the management of customer relationships, considering customers as their most valuable assets due to their pivotal role in generating revenue. Consequently, companies are increasingly cognizant of the need to invest efforts in both retaining existing customers and acquiring new ones. In this context, the concept of churners emerges—individuals who frequently switch service providers or products for various reasons. Recognizing the significance of accurately predicting consumer behavior and identifying trends in customer attrition, businesses aim to mitigate customer turnover by employing predictive analytics techniques. Central to this effort is the practice of churn prediction, which involves employing binary classification methods to differentiate between customers who are likely to churn and those who are not. By conducting a comparative analysis of various machine learning algorithms for customer churn prediction, this study seeks to provide insights into the effectiveness of different methodologies, namely Sequential, Random Forest, and Decision Tree Classifier models. Through this analysis, we aim to identify the most suitable approach for predicting customer churn, thereby assisting businesses in devising proactive strategies to retain customers and sustain long-term profitability.

In their study cited as Ahmad et al. (2019) [4], Damandeep Singh, Vansh, and Dr. M. Kanchana meticulously employed clean data for their churn prediction model, aiming to eliminate any irregularities that could compromise the model's accuracy. Through a rigorous selection process, only features with a proven impact on customer attrition were considered, using criteria such as benefit data and qualification rating filters. This meticulous approach not only enhances the model's predictive power but also ensures its relevance to real-world scenarios. By leveraging Customer Relationship Management (CRM) tools, the model not only boosts operational efficiency but also enables targeted marketing strategies and timely interventions based on predictive insights derived from customer data. The study underscores the critical importance of accurately predicting churn, as it empowers businesses to proactively address customer attrition and optimize marketing campaigns. The model's ability to accurately predict market behavior based on past customer conduct underscores its efficacy in guiding strategic decision-making. Specifically tailored for the telecom industry, this model analyzes consumer information to forecast churn probabilities, thereby providing invaluable insights for business sustainability and growth.

In Singh et al.'s survey (2021) [8], conducted by Rehan Ullah Khan, Mohamed Tahar Ben Othman, and Amir Ali Mustafa Qamar, the escalating competition within the telecommunications sector has rendered the prediction of customer churn across diverse service providers increasingly challenging. Acknowledging the cost-effectiveness of retaining existing clients over acquiring new ones and the higher revenue contribution from long-term customers, businesses are compelled to devise strategies to mitigate client turnover. Machine learning, with its capacity to discern patterns and extrapolate them for predicting future occurrences, has emerged as a pivotal tool in this endeavor, offering a potent paradigm for data mining.

In Almufadi's study (2019) [9], Kolla Bhanu Prakash explored the feasibility of employing deep learning, a subset of machine learning, for image recognition tasks. By utilizing a neural network model, this research delved into the realm of deep learning, particularly focusing on its application in picture identification. Specifically, the study examined the efficacy of neural networks in discerning the intricate characteristics of three-dimensional coordinates, which are instrumental in soil categorization and parameter assessment. The findings suggest the potential benefits of leveraging Artificial Neural Networks (ANN) in soil classification tasks, indicating a promising avenue for utilizing AI-driven approaches in environmental science and agriculture.

In Kalyani's presentation cited in Kriti's statement [10], the importance of comprehending the reasons behind client turnover is emphasized, highlighting factors such as pricing, benefits, and the duration of the customer-business relationship as common elements across various market sectors. Previous research has employed regression modeling to delve into consumer behavior analysis, while clustering techniques have been utilized to segment client groups with similar behaviors. One notable study, "A Review on Customer Churn Prediction in Telecommunication Using Data Mining Techniques" by S. Babu and Dr. N. R. Ananthanarayanan, focuses specifically on the telecommunications industry, categorizing the factors influencing client churn.

In Ghosh's study [11], attention is directed to the banking industry, where forecasting client churn based on past credit behavior is crucial for risk management. Particularly, the focus lies on predicting whether a client would default in the upcoming quarter, enabling proactive measures to be taken. Meanwhile, in Peddarapu Rama Krishna and Pothuraju Rajarajeswari's work, the use of the EfficientNetB6 Algorithm in a deep learning approach is proposed, demonstrating its efficacy in non-invasive computer vision-assisted clinical diagnosis in medicine.

Additionally, [12] Mustafa, Ling, and Razak's research underscores the significance of customer turnover in the telecommunications sector, where the churn rate serves as a key metric for assessing market competitiveness. With the telecom industry losing a substantial percentage of clients daily, understanding customer preferences and offering competitive pricing and subscription options are essential for retaining market share.

II. Methodology

Machine learning and deep learning are both subfields of artificial intelligence, yet they differ in their approach and complexity. While machine learning encompasses a broader range of techniques for teaching computers to learn from data, deep learning specifically focuses on neural networks with multiple layers, allowing for more intricate pattern recognition. Table 1 elaborates on the crucial differences between machine learning and deep learning.

A. Role of Deep Learning

The role of deep learning algorithms, especially Artificial Neural Networks (ANNs), in analyzing datasets for churn patterns has revolutionized various fields, particularly in multiple industries where early detection of customer churn is crucial for ensuring customer retention. Here's an elaborate discussion of key aspects:

B. Complex Pattern Recognition

- Deep learning algorithms, especially neural networks with multiple hidden layers, excel at recognizing intricate patterns within large and complex datasets.
- In customer churn prediction, these patterns may include subtle changes in user behavior, transaction history, or engagement levels that signal an increased likelihood of churn.
- By recognizing complex patterns, deep learning models can accurately predict potential churn events before they occur.

C. Temporal Dynamics

- Deep learning architectures like Artificial Neural Networks (ANNs) and Long Short-Term Memory Networks (LSTMs) are well-suited for capturing temporal dependencies in sequential data.
- Customer behavior often evolves over time, and deep learning models can analyze historical patterns to predict future churn.
- These models consider the sequence of events, such as changes in purchase frequency or engagement levels, to make accurate churn predictions.

D. Advancement in Churn Prediction

Advancements in customer churn prediction using deep learning have significantly improved the accuracy, efficiency, and interpretability of predictive models. Here are some key advancements:

E. Advanced Model Architectures

- Deep learning architectures such as Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Gated Recurrent Units (GRUs) have been extensively applied to capture temporal dependencies in sequential customer data, enabling more accurate churn predictions over time.
- Attention mechanisms and transformer architectures have enhanced model performance by focusing on relevant parts of the input sequence, allowing for more precise predictions based on informative features.

F. Model specifications

1) Sequential Model: The sequential model facilitates the specification of a neural network in a linear manner, progressing from input to output by traversing through a sequence of neural layers, each interconnected and arranged sequentially [13].

TABLE I: Comparison between Machine Learning and Deep Learning Approaches

Aspect	Machine Learning	Deep Learning
Complex Pattern Recognition	Capable of capturing simple to moderate patterns	Excels in capturing intricate and complex patterns
Feature Engineering	Manual feature engineering required	Automated feature learning from raw data
Scalability to Big Data	Less scalable compared to deep learning	Highly scalable and efficient for big datasets
Real-time Intervention	Limited capability for real-time interventions	Facilitates real-time interventions and personalization
Temporal Dynamics Modeling	Limited ability to capture temporal dependencies	Able to model temporal dependencies effectively

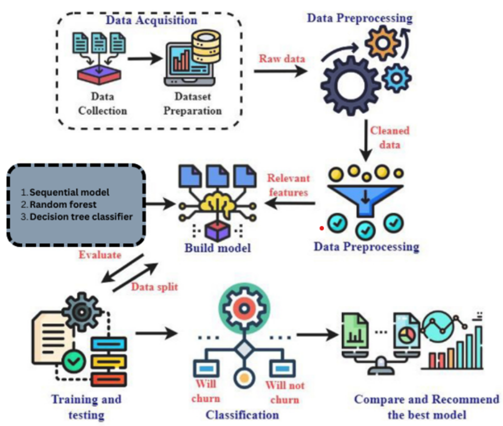

Fig. 1: Workflow Diagram

2) Random Forest Algorithm: In a Random Forest, each decision tree is constructed based on a random subset of features and a bootstrapped sample of the dataset, ensuring diversity among the trees. The ensemble nature of Random Forest reduces overfitting and increases robustness by averaging or voting on the predictions of individual trees [14]. The aggregation process involves combining these diverse predictions to reach a final output. Through a process known as voting, each tree ”votes” on the predicted class for a given input, and the most commonly predicted class becomes the final prediction. This approach enhances the model's accuracy and generalization ability, making Random Forest a powerful tool for classification tasks in various domains, from finance to healthcare and beyond [15].

3) Decision Tree Classifier: A decision tree classifier partitions the feature space into subsets based on input features, aiming to maximize class purity [16]. It iteratively selects the best feature to split the data until fully classified, resulting in a tree-like structure with leaves representing final class predictions. Decision trees offer interpretability and simplicity, visualizing factors driving churn in customer churn prediction [17]. However, they may overfit complex datasets, mitigated by techniques like pruning or ensemble methods. Despite limitations, decision trees provide valuable insights into churn drivers and complement other machine learning approaches in building accurate prediction models [18].

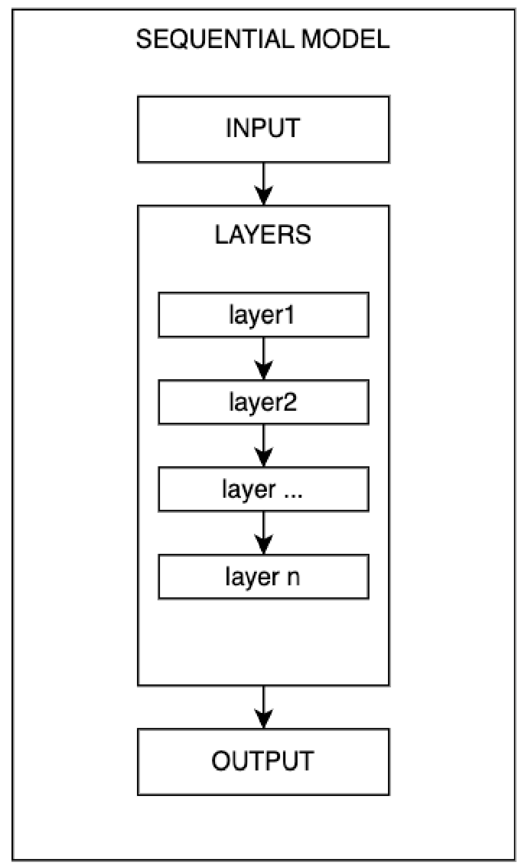

Fig. 2: Sequential model

G. Objective of Deep Learning in Customer Churn Prediction

The objective of deep learning in customer churn prediction involves leveraging advanced computational techniques to accurately anticipate and mitigate customer attrition. Here are some subtopics and points under each:

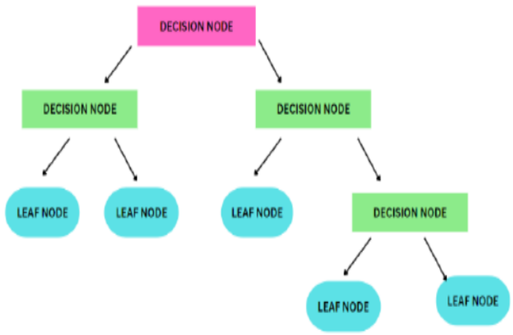

Fig. 3: n-D-Trees-based classification

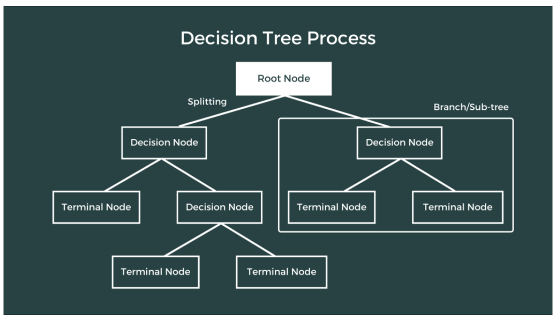

Fig. 4: Decision tree classifier

1) Improved Prediction Accuracy:

- Deep learning aims to enhance the accuracy of churn prediction models by capturing intricate patterns and relationships within large and complex datasets.
- Substantial improvement in prediction accuracy enables businesses to identify potential churners more effectively and allocate resources towards targeted retention strategies.

2) Automated Feature Learning:

- Deep learning techniques automate the process of feature extraction from raw data, eliminating the need for manual feature engineering.
- By automatically learning hierarchical representations of data, deep learning models can uncover hidden insights and patterns that may not be apparent through traditional feature engineering approaches.

H. Scalability to Big Data

- Deep learning techniques are scalable and capable of processing large volumes of data efficiently.
- Scalability to big data enables businesses to analyze vast amounts of customer data, including transaction history, demographic information, and behavioral patterns, to make informed churn predictions.

III. Results and Discussion

A. Data Collection

We collected the data from Kaggle, comprising datasets from two different fields.

1) Dataset of Telecom Industries: The dataset is provided in CSV format, containing a total of 21 features and 7043 records. Dummy values have been dropped from the dataset, duplicate records are not considered and are removed beforehand before passing to the model, and NULL values are replaced with aggregate functionality (average of the column attribute). Figure 5 shows a sample of the telecom dataset used in our program.

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection
1374	8999-YPYBV	Male	0	Yes	Yes	45	Yes	No	DSL	Yes	...	Yes
4794	0366-NQSHS	Male	0	No	No	2	Yes	No	No	No internet service	...	No internet service
5284	3317-HRTNN	Female	1	No	No	5	No	No phone service	DSL	No	...	No
3915	3115-JPJDD	Female	0	Yes	No	59	Yes	No	Fiber optic	Yes	...	Yes
6720	9530-GRMJG	Male	0	Yes	Yes	70	Yes	Yes	DSL	No	...	Yes

5 rows × 21 columns

Fig. 5: Sample of the telecom dataset

We have converted the object data types present in the dataset into signed and unsigned integers. We segregate the column values (created dummy columns) to convert the data types of the features. Figure 6 shows the datatypes of the dataset and figure 7 changed datatypes which we changed to optimize the performance.

B. Dataset of banking sector

The dataset is in CSV format. In this dataset, we have a total of 18 features and 10000 records. Dummy values are dropped from the dataset; duplicate records are not considered and are removed before passing to the model, and NULL values are replaced with the aggregate functionality (Average of the column attribute). Figure 8 shows the banking sector dataset which we are using in our program.

We have converted the object data types present in the dataset into signed and unsigned integers. We segregate the column values (created dummy columns) to convert the data types of the features.

C. Code Implementation

1) Python: Python is the go-to language for deep learning due to its readable syntax and rich libraries like Scikit-learn, TensorFlow, and PyTorch. Its versatility, open-source nature, and robust community support make it ideal for developing, training, and deploying ML models, fostering innovation in the field.

2) TensorFlow: TensorFlow is an open-source machine-learning library developed by Google. Widely used in AI research and production, it facilitates the creation, training, and deployment of machine learning models, particularly neural networks. TensorFlow offers a flexible platform for various ML tasks, with broad community support and compatibility across different devices.

3) Matplotlib: Matplotlib is a popular 2D plotting library for the Python programming language. It enables the creation of static, animated, and interactive visualizations in Python. Matplotlib is widely used for data visualization in fields such as data science, machine learning, and scientific research, providing versatile and customizable plotting capabilities.

```
gender              object
SeniorCitizen        int64
Partner             object
Dependents          object
tenure               int64
PhoneService        object
MultipleLines       object
InternetService     object
OnlineSecurity      object
OnlineBackup        object
DeviceProtection    object
TechSupport         object
StreamingTV         object
StreamingMovies     object
Contract            object
PaperlessBilling    object
PaymentMethod       object
MonthlyCharges     float64
TotalCharges        object
Churn               object
dtype: object
```

(a) Data types

```
gender                                       int64
SeniorCitizen                                int64
Partner                                      int64
Dependents                                   int64
tenure                                       int64
PhoneService                                 int64
MultipleLines                                int64
OnlineSecurity                               int64
OnlineBackup                                 int64
DeviceProtection                             int64
TechSupport                                  int64
StreamingTV                                  int64
StreamingMovies                              int64
PaperlessBilling                             int64
MonthlyCharges                             float64
TotalCharges                               float64
Churn                                        int64
InternetService_DSL                          uint8
InternetService_Fiber optic                  uint8
InternetService_No                           uint8
Contract_Month-to-month                      uint8
Contract_One year                            uint8
Contract_Two year                            uint8
PaymentMethod_Bank transfer (automatic)      uint8
PaymentMethod_Credit card (automatic)        uint8
PaymentMethod_Electronic check               uint8
PaymentMethod_Mailed check                   uint8
dtype: object
```

(b) Changed data types

Fig. 6: Comparison of data types

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember
8490	8491	15585985	Wang	746	France	Male	48	5	165282.42	1	1	0
7507	7508	15642001	Lorenzen	576	Germany	Male	44	9	119530.52	1	1	0
3799	3800	15655341	Chinagorom	458	Spain	Female	35	5	166492.48	1	1	0
3826	3827	15751774	Monnier	774	France	Male	76	4	112510.89	1	1	1
290	291	15652266	Chidiebele	703	Germany	Male	42	9	63227.00	1	0	1

Fig. 7: Sample image of banking sector dataset

```
CreditScore              int64
Gender                   int64
Age                      int64
Tenure                   int64
NumOfProducts            int64
HasCrCard                int64
IsActiveMember           int64
EstimatedSalary        float64
Churn                    int64
Complain                 int64
Satisfaction Score       int64
Point Earned             int64
Card Type_DIAMOND        uint8
Card Type_GOLD           uint8
Card Type_PLATINUM       uint8
Card Type_SILVER         uint8
Geography_France         uint8
Geography_Germany        uint8
Geography_Spain          uint8
dtype: object
```

Fig. 8: Data types of the dataset

4) Keras: Keras is an open-source high-level neural network API written in Python. Initially developed as an independent project, it has been integrated into TensorFlow as its official high-level API. Keras simplifies the process of building, training, and deploying deep learning models, making it accessible to both beginners and experienced practitioners. Here we have used Keras to create a sequential model, extract the feature from the dataset, and predict customer churn.

5) Optimizer: In Artificial Neural Networks(ANNs), common optimizers like Adagrad, Adam, SGD, and RMSprop adjust model parameters during training to minimize the loss function. These optimizers, available in Keras, play a crucial role in determining convergence speed and final performance, with choices based on task characteristics and experimentation.

6) Data Training: Training in machine learning involves optimizing a model's parameters using a dataset. For Artificial Neural Networks (ANNs), this process includes feeding input data, calculating the loss, and adjusting weights with an optimizer. Iterative epochs refine the model, enabling it to make accurate predictions on new, unseen data.

7) Handling imbalance dataset: Handling imbalanced datasets in customer churn prediction is crucial for building robust and accurate models. An imbalance occurs when the number of churn instances is significantly lower than non-churn instances, leading to biased model performance. Techniques such as oversampling, under-sampling, and Synthetic Minority Over-sampling Technique (SMOTE) can address this issue by balancing the class distribution in the dataset. By generating synthetic samples or adjusting the sample weights, these methods ensure that the model learns from both positive and negative examples, improving its ability to predict churn accurately and effectively mitigate customer attrition.

The figure depicts a comparison of churn rates relative to customer tenure, where green bars represent retained cus-

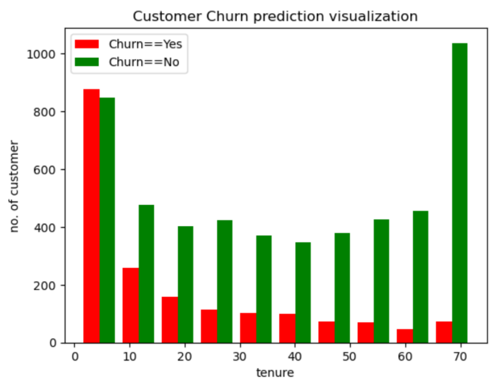

Fig. 9: EDA analysis of Churn and Tenure

tomers and red bars denote churned customers. The x-axis likely represents different tenure periods, while the y-axis displays the corresponding churn rates. The green bars illustrate lower churn rates among customers with longer tenures, indicating higher retention levels. Conversely, the red bars suggest higher churn rates among customers with shorter tenures, indicating a propensity to discontinue services earlier in their tenure. This visualization underscores the importance of customer retention strategies, particularly in extending the tenure of customers to mitigate churn and enhance overall business stability.

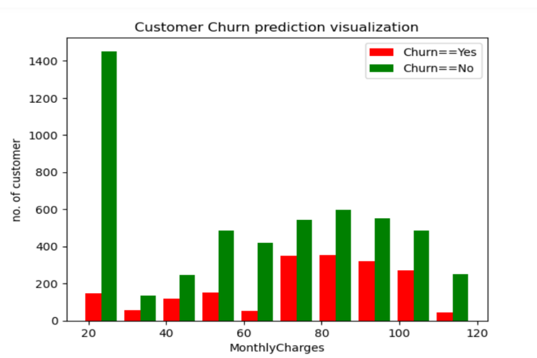

Fig. 10: EDA analysis of churn and Monthly Charge

The figure provides insights into the relationship between monthly charges and customer retention, with the x-axis likely representing different monthly charge brackets and the y-axis showing the corresponding churn rates. The visualization may reveal trends such as higher churn rates among customers paying lower monthly charges, suggesting a potential sensitivity to pricing. Conversely, lower churn rates may be observed among customers in higher charge brackets, indicating a stronger commitment or satisfaction level associated with premium services. This analysis highlights the importance of pricing strategies in customer retention efforts, suggesting that optimizing pricing tiers or offering value-added services may positively influence retention rates and overall customer loyalty.

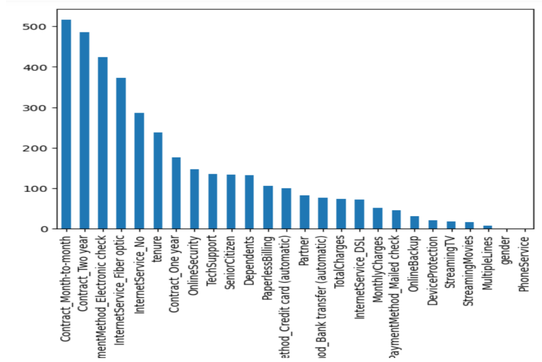

Fig. 11: output of feature selection algorithm

The figure presents the feature importance rankings for churn prediction within the dataset, arranged in descending order. Each feature's importance is likely determined through a method such as Gini importance or permutation importance, reflecting its contribution to the predictive performance of the model. Features with higher importance scores are deemed more influential in predicting churn behavior, while those with lower scores have relatively less impact. Understanding these feature rankings enables prioritization in feature selection, model optimization, and decision-making processes. By focusing on the most important features, businesses can develop targeted strategies to address key factors driving customer churn, thereby enhancing retention efforts and maximizing customer lifetime value.

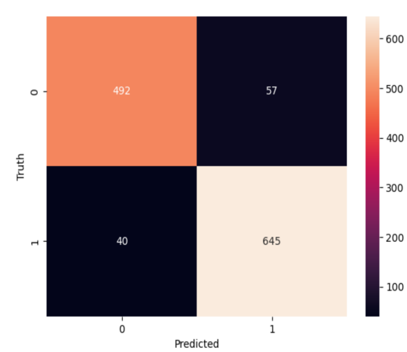

Fig. 12: Heatmap of sequential model confusion matrix

The figure illustrates a heatmap of the confusion matrix generated by a sequential model, where the rows represent the actual values and the columns represent the predicted values. Each cell in the heatmap corresponds to the frequency or proportion of instances falling into a particular combination of actual and predicted classes. This visualization provides a comprehensive overview of the model's performance across different classes, highlighting areas of accurate predictions (diagonal elements) as well as misclassifications (off-diagonal elements). Analyzing the heatmap enables insights into the model's strengths and weaknesses, such as its ability to correctly identify certain classes or potential areas for improvement, guiding further optimization or refinement of the predictive model.

	Sequential Model	Random Forest	Decision Tree Classifier
Precision	0.94	0.92	0.90
Recall	0.94	0.92	0.90
F1-Score	0.94	0.92	0.90
Accuracy	0.94	0.92	0.90

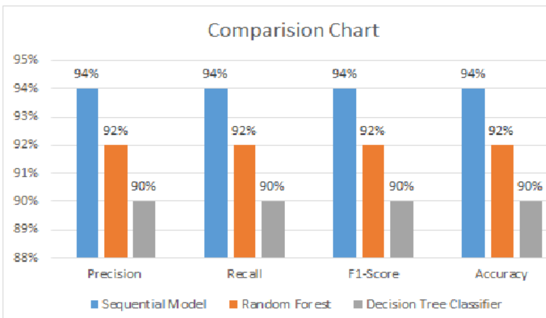

Fig. 13: Comparision Chart

The chart displays a comparative analysis of three machine learning models: Sequential Model, Random Forest, and Decision Tree Classifier. The models are evaluated based on four performance metrics: Precision, Recall, F1-Score, and Accuracy. The values for each metric are represented as percentages. This comparative analysis highlights that while all three models are useful for customer churn prediction, the Sequential Model stands out as the most reliable and accurate option based on the given data.

IV. Conclusion

This comparative study investigated the effectiveness of Sequential models, Random Forest, and Decision Tree classifiers in predicting customer churn. Our analysis demonstrated that the Sequential model outperforms the other two algorithms with an accuracy of 94%, while Random Forest and Decision Tree classifiers achieved 92% and 91%, respectively.

The Sequential model's superior performance can be attributed to its ability to capture temporal dependencies in sequential data, providing deeper insights into customer behavior patterns over time. However, Random Forest and Decision Tree classifiers also proved to be valuable due to their simplicity, interpretability, and relatively high accuracy.

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